Milestone 1 - Complete Topic Guide

LTM AI Training | Days 1-11 | Milestone Assessment 1 + Pre-CEFR on 29 Oct 2026 (Day 12)

1. Key dates

Date	Day	Event
12 Oct 2026	1	Training starts
21 Oct 2026	6	Weekly Assessment 1
27 Oct 2026	10	Mock Assessment 1
28 Oct 2026	11	Mock Assessment 2 (last day of new content)
29 Oct 2026	12	**Milestone Assessment 1 + Pre-CEFR**
3 Nov 2026	15	Milestone Assessment 1 (2nd attempt)

Note: the syllabus only lists 'Pre-CEFR' on the milestone day without details. Ask your trainer or batch coordinator what it covers.

2. Day-by-day syllabus

Day	Date	Session 1	Session 2	Session 3
1	12 Oct	Environment setup (VS Code, Anaconda, Jupyter, venv, conda)	Basic syntax: variables, data types, operators	Control flow: if-else, for, while
2	13 Oct	Data structures: lists, tuples, dictionaries, sets	Functions: args, *args, **kwargs, scope	OOP: classes, objects, attributes, methods, constructors
3	14 Oct	OOP: inheritance, polymorphism, encapsulation	Modules and packages; file I/O (text, CSV, JSON)	Error handling, debugging, documentation
4	15 Oct	Iterators, generators, list/dict comprehensions	Decorators	NumPy: arrays, broadcasting, vectorised computation
5	16 Oct	Pandas: DataFrames, Series, data loading	Pandas: cleaning, manipulation, aggregation, merging	SQL Fundamentals Part 1
6	21 Oct	SQL Fundamentals Part 2: complex scenarios	Database connectivity and CRUD operations	Vector databases: Chroma, PGVector, FAISS **+ Weekly Assessment 1**
7	22 Oct	Version control with Git and GitHub	API calling and response processing; PDF extraction and OCR	Streamlit: building POC apps, deployment, best practices
8	23 Oct	FastAPI concepts and setup: HTTP, ASGI vs WSGI, uvicorn, /docs	Basic endpoints, GET/POST, status codes, headers; Postman/curl testing	In-memory CRUD, Pydantic models, response models, PUT/DELETE, path and query params
9	26 Oct	Error signalling: 404/400/422, custom headers, response model design	Database fundamentals: ORM, SQLAlchemy/SQLModel, models and schemas	CRUD with ORM, relationships, Alembic, PostgreSQL/SQLite connection and pooling
10	27 Oct	Middleware and CORS: architecture, request/response lifecycle	Redis basics, caching strategies, custom middleware, performance testing	Security fundamentals: authn vs authz, password hashing, JWT **+ Mock Assessment 1**
11	28 Oct	Token-based auth, JWT, OAuth2, securing endpoints, login/signup, protected routes, RBAC	Testing strategies: unit vs integration, pytest setup, mocking and fixtures	Dependency injection, modular design, router organisation, API tests, async endpoints, logging **+ Mock Assessment 2**

3. Coding checklist: Python, OOP, NumPy, Pandas (highest priority)

Tick each item only after you can write it from a blank file without help.

Done	A. Python basics (Days 1-2)
	Variables, data types, type conversion, operators, input/output
	Strings: indexing, slicing, methods (split, join, strip, replace), f-strings
	Conditionals; for and while loops; range; break/continue; nested loops
	Lists: methods, slicing, sorting, nested lists
	Tuples and sets: uniqueness, set operations, when to use which
	Dictionaries: counting, grouping, nested dicts, a list of dicts
	Functions: parameters, return, default values, scope, *args, **kwargs

Done	B. Advanced Python (Days 3-4)
	lambda, map, filter, sorted with key
	List and dictionary comprehensions
	Iterators and generators (yield) and why they save memory
	Decorators: writing a simple one (e.g. a timer)
	File I/O: read and write text, CSV, JSON
	try/except/else/finally, raising exceptions, custom exceptions
	Modules and packages, imports; collections, datetime, random, os
	Docstrings, basic debugging

Done	C. OOP (Days 2-3)
	Classes, objects, __init__, self, methods, attributes
	__str__ and __repr__; class vs instance variables
	@classmethod and @staticmethod
	Encapsulation: private attributes, @property
	Inheritance, super(), method overriding
	Polymorphism and abstract classes (abc)
	Mini-systems: bank, library, inventory with error handling

Done	D. NumPy (Day 4)
	Creating arrays; shape, dtype, reshape
	Indexing, slicing, boolean masking
	Vectorised math and broadcasting
	Aggregation with axis (sum, mean, max)
	where, sort, unique, concatenate, random numbers
	Replacing loops with vectorised code

Done	E. Pandas (Day 5)
	Series and DataFrame; loading CSV/JSON; head, info, describe
	loc and iloc; selecting columns; filtering with conditions
	Missing values (isna, fillna, dropna), duplicates, dtypes, renaming
	apply, map, creating new columns, sorting, string and date methods
	groupby, agg, pivot_table, value_counts
	merge, join, concat
	Answering business questions from a real CSV

Done	F. SQL (Days 5-6)
	SELECT, WHERE, ORDER BY, LIMIT, DISTINCT
	GROUP BY, HAVING, aggregate functions
	Joins (INNER, LEFT, RIGHT) and subqueries
	Complex scenarios: multi-table queries, filtering with conditions
	CRUD: INSERT, UPDATE, DELETE; connecting to a database from Python

4. Backend and tools checklist (Days 6-11)

These are part of Days 1-11, so expect questions on them in the milestone.

Done	G. Tools and APIs (Days 6-7)
	Vector databases: what embeddings are, Chroma vs PGVector vs FAISS, basic similarity search
	Git and GitHub: init, add, commit, branch, merge, push, pull, pull request
	Calling APIs with requests; handling JSON responses and errors
	PDF text extraction and OCR basics
	Streamlit: building a small proof-of-concept app and deploying it

Done	H. FastAPI (Days 8-9)
	HTTP basics, ASGI vs WSGI, running with uvicorn, the /docs page
	GET/POST endpoints, path and query parameters, headers, status codes
	Testing with Postman or curl
	Pydantic models and response models; PUT and DELETE
	Error signalling: 404, 400, 422, custom headers
	ORM with SQLAlchemy or SQLModel; models vs schemas; relationships
	Alembic migrations; PostgreSQL/SQLite connection and pooling

Done	I. Advanced backend (Days 10-11)
	Middleware and CORS: request/response lifecycle, custom middleware
	Redis basics and caching strategies; performance testing
	Authentication vs authorization; password hashing
	JWT, OAuth2, login/signup, protected routes, RBAC
	Testing: unit vs integration, pytest setup, mocking and fixtures
	Dependency injection, modular design, router organisation
	API tests, async endpoints, logging

5. Suggested priority

- **Highest:** Python basics, OOP, NumPy, Pandas, SQL. These are your coding focus and the foundation of everything else.
- **High:** FastAPI basics, Pydantic, error handling, ORM CRUD. You can reuse your Express/REST knowledge here.
- **Medium:** Auth and JWT, middleware and CORS, testing with pytest, Git, API calling.
- **Understand the idea:** Redis caching, vector databases, Streamlit, OCR, Alembic.

6. How to use this guide

- Work through sections A to F first, then G to I.
- For each item, rebuild the code from a blank file before ticking it.
- Keep a mistakes log (error, cause, fix) and review it before 27-29 Oct.
- Take mixed timed tests on 27-28 Oct, then redo anything you got wrong.
- Topics are taken from the training planner; confirm the exact assessment scope with your trainer.